

NEUR-503 - Introduction to Computational Neuroscience

Modeling a Cortex-Basal Ganglia-Cerebellum Architecture Based on Doya (1999, 2000):

Implementation, Experimentation, and Critique

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1 Appendix

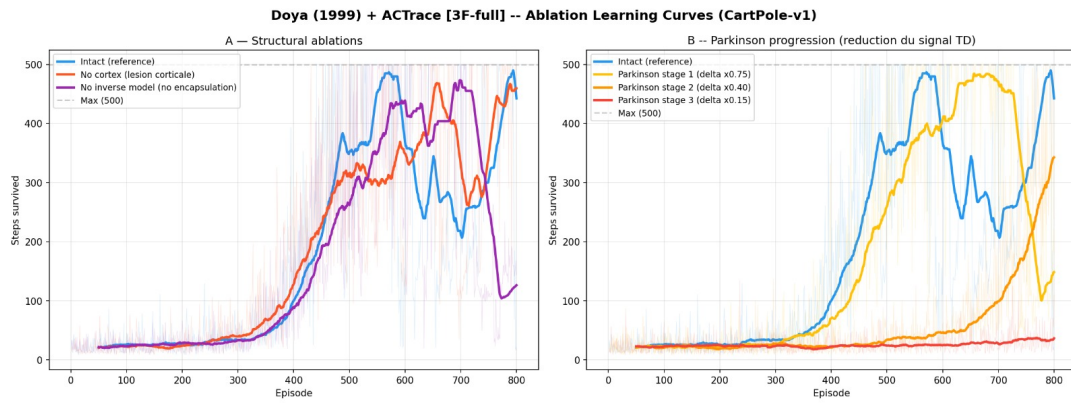


Figure 1: Learning curves in reactive mode under the conditions: Intact (reference=3F-Full), No cortex, No inverse model and Parkinson (dopamine $\times 0.75$, $\times 0.40$, $\times 0.15$). The training was performed on the Cartpole environment for 800 epochs. These graphs correspond to the results observed from ablation experiments n°1, n°3 and the Parkinson simulation.

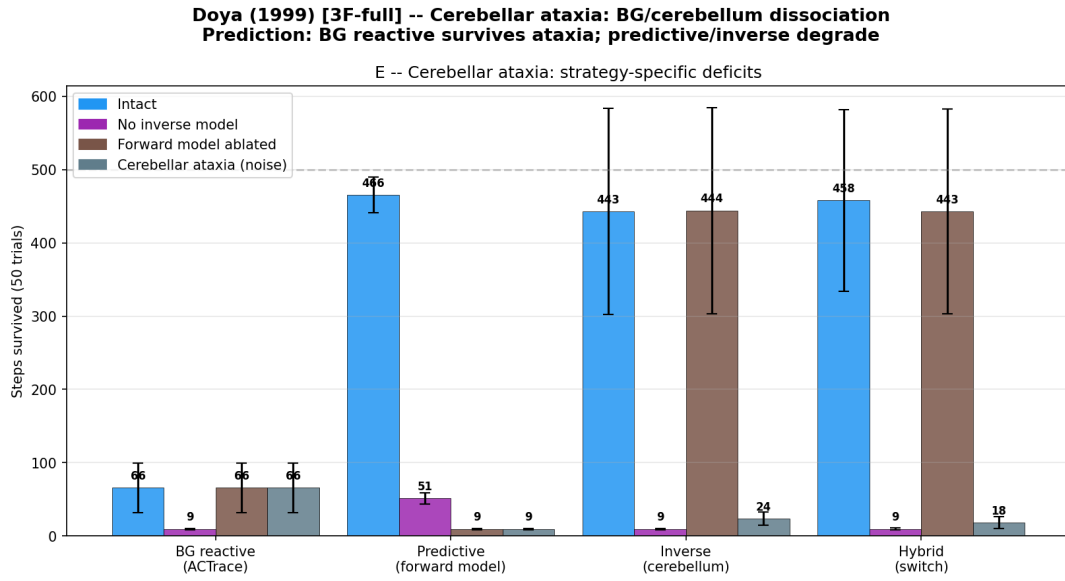


Figure 2: Tests (50 trials) conducted after training in **predictive** mode over 800 epochs. The tested modalities are: Reference (3F-Full), Forward model ablated, Cerebellar ataxia and no inverse model. All inference strategies are tested. These graphs correspond to the results observed during ablation experiment n°2 n°3 and the cerebellar ataxia simulation.

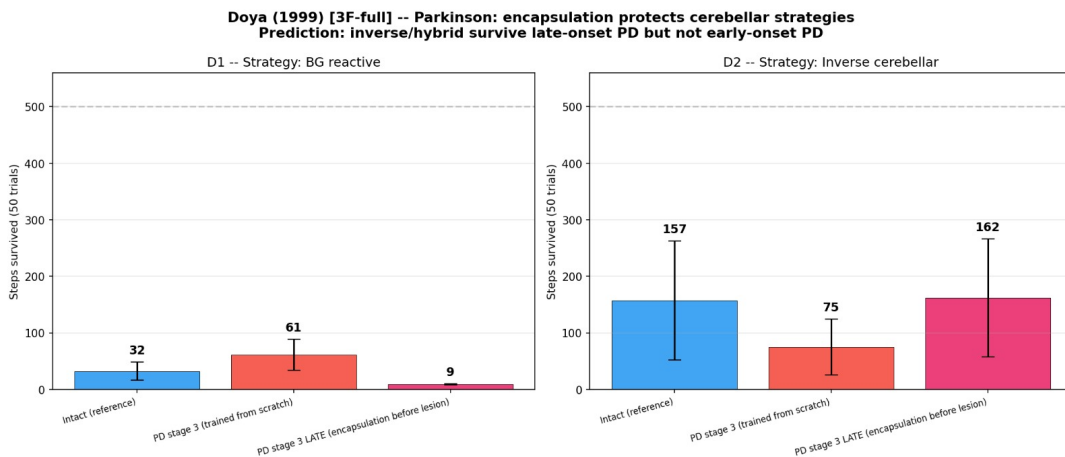


Figure 3: Tests (50 trials) conducted after training in reactive mode over 800 epochs. The tested modalities are: Reference (3F-Full), Parkinson (advanced stage, $\delta = 0.15$ where in one condition training was conducted after the lesion (PD stage 3 LATE) and in the other before (PD stage 3). These graphs correspond to the results observed during the Parkinson simulation. The tested modalities are Basal ganglia alone in reactive mode and Inverse cerebellum.